

S M (Rifat) Rafiuddin

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📍 716 N Husband Street, Stillwater, OK, USA.

RESEARCH PROFILE

Ph.D. researcher in NLP and efficient, trustworthy language models, with interests in LLM reasoning, inference optimization, and interpretability.

EDUCATION

- **Oklahoma State University** Stillwater, OK, USA
Ph.D. in Computer Science Aug 2022 – Jul 2027 (Expected)
- **Rajshahi University of Engineering and Technology (RUET)** Rajshahi, Bangladesh
B.Sc. in Computer Science and Engineering Jan 2012 – Oct 2016

TECHNICAL SKILLS

- **LLMs & NLP:** Transformers, LoRA/PEFT, token masking, chain-of-thought reasoning, ABSA/MABSA, NLTK
- **Deep Learning & ML:** PyTorch, TensorFlow, Scikit-learn, NumPy, pandas, Matplotlib
- **Optimization & Systems:** Efficient inference, memory/throughput profiling, Docker, Linux, Git
- **Programming Languages:** Python, C, C++, Java
- **Data & Cloud:** Amazon AWS, Hadoop, Apache Spark, MySQL, Oracle, PL/SQL

EXPERIENCE

- **Oklahoma State University, rAison Lab** Stillwater, OK, USA
Graduate Researcher, Reasoning and Artificial Intelligence Jan 2025 – Present
 - **Sentiment reasoning (ABSA/MABSA):** Developed multimodal sentiment reasoning models with aspect-aware fusion and attention alignment (text+image), and ran ablations to identify the most critical components.
 - **Causal & robust multimodal reasoning:** Investigated robustness, causal-style alignment in multimodal learning (e.g., cross-modality alignment + feedback for real-time emotion understanding) under missing, noisy modalities.
 - **Masking & optimization:** Built masking-based training, inference methods (contextual, syntax-guided masking; token-retention gating) to improve efficiency and memory, throughput, with fully reproducible pipelines and profiling.
- **Oklahoma State University** Stillwater, OK, USA
Graduate Teaching Assistant Aug 2022 – Present
 - **Course Assistance:** Assisted in teaching 750+ students in Computer Security, OS Design, and Algorithm Analysis.
 - **Mentorship:** Led labs, graded assignments, and mentored students in systems and algorithmic courses.
 - **Content Design:** Designed assignments and exams to reinforce hands-on and conceptual understanding.
- **University of Asia Pacific** Dhaka, Bangladesh
Lecturer Oct 2018 – Jul 2022
 - **Course Instruction:** Delivered core CS courses and labs using interactive, applied teaching strategies.
 - **Programming Team Coaching:** Coached RUET IUPC 2019 programming team to notable competitive success.
 - **Curriculum Development:** Contributed to OBE-based curriculum enhancement through IQAC workshops.
- **Uttara University** Dhaka, Bangladesh
Lecturer Feb 2017 – Oct 2018
 - **Teaching:** Taught Discrete Math, Java, C++, Algorithms, and OS; ran labs.
 - **Assessment:** Graded with feedback; supported exams (questions, moderation, results).

SELECTED PUBLICATIONS (MOST RECENT FIRST)

- Rafiuddin, R., 2026. **MaskLoRA: Low-Rank Subspace-Induced Token Masking for Efficient and Faithful Language Models.** In *Findings of the Association for Computational Linguistics: EACL 2026*. (CORE A Conference)
- Rafiuddin, S.M. and Khan, M.N., 2026. **Emergent Discrete Controller Modules for Symbolic Planning in Transformers.** In *International Conference on Learning Representations (ICLR 2026)*. (CORE A* Conference)
- Rafiuddin, S.M. and Khan, M.N., 2025, December. **A Formal Analysis of Chain-of-Thought Prompting via Turing Reductions.** In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AAACL 2025)* (pp. 87–98), Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics. (CORE B Conference)
- Rafiuddin, S.M., 2025. **Edu-EmotionNet: Cross-Modality Attention Alignment with Temporal Feedback Loops.** In *International Conference on Machine Learning and Applications (ICMLA 2025)*, Boca Raton, FL, USA, Dec 3–5, 2025. doi:10.48550/arXiv.2510.08802 (CORE C Conference)
- Rafiuddin, S.M. and Khan, M.N., 2025. **Learning What to Remember: Adaptive Probabilistic Memory Retention for Memory-Efficient Language Models.** In *Findings of the Association for Computational Linguistics: EMNLP 2025* (pp. 3969–3981), Suzhou, China. Association for Computational Linguistics. doi:10.18653/v1/2025.findings-emnlp.212 (CORE A* Conference)

- Rafiuddin, S.M., Kamal, S., Rakib, M., Bagavathi, A. and Sen, A., 2025. **AdaptiSent: Context-Aware Adaptive Attention for Multimodal Aspect-Based Sentiment Analysis.**: In *International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. doi:10.1007/978-3-032-13513-1_11 (CORE B Conference)
- Rafiuddin, S.M., Rakib, M., Kamal, S. and Bagavathi, A., 2024, April. **Exploiting Adaptive Contextual Masking for Aspect-Based Sentiment Analysis.**: In *Advances in Knowledge Discovery and Data Mining (PAKDD 2024)*, Lecture Notes in Computer Science (LNAI), vol. 14650, pp. 147–159. Springer, Singapore. doi:10.1007/978-981-97-2266-2_12 (Acceptance Rate: 18.47%) (CORE B Conference)

PROJECTS

- **Image Embedding and Classification (Vision, Deep Learning)**: Implemented Xception-based pipeline to extract and visualize high-dimensional image embeddings with downstream classification; built clean training/evaluation scripts and TensorBoard 2D/3D demos. Tech: TensorFlow, NumPy, Matplotlib (2021). [GitHub]
- **GAN-based Data Augmentation for Bangla Characters (Generative Models, Low-Resource Vision)**: Applied adaptive discriminator augmentation to stabilize GAN training on limited data; produced robust character/numeral synthesis with visual demos and reproducible configs. Tech: PyTorch, GANs, OpenCV (2021). [GitHub]
- **Breast Cancer Detection with Deep Learning (Medical Imaging, CNNs)**: Built and compared Inception, VGG16, MobileNet, and Transformer-based models for IDC detection in histopathology images; constructed an end-to-end data pipeline and evaluation on Kaggle IDC. Tech: TensorFlow, Keras, NumPy (2021). [GitHub]
- **GO-CART, 3D Unity Game (Real-Time Graphics & Physics)**: Designed a Unity/C# 3D racer with real-time gameplay loop: WASD controls, third-person camera, physics-based handling, collision detection, scoring and game-over logic; tuned for smooth interaction. Tech: Unity, C# (2021). [GitHub]
- **Adaptive Blockchain with Dynamic Difficulty & SJF (Systems, Queuing)**: Simulated blockchain under high-load; integrated dynamic difficulty control and Shortest-Job-First prioritization via min-heap to improve throughput and reduce queue length/wait time; course project (OSU CS5113). Tech: Python, Priority Queues (2024). [GitHub]

WORK AUTHORIZATION

CPT-eligible for **Summer 2027 internship** (May–Aug); at least one academic term remaining after internship.

HONORS AND AWARDS

- **AIRS Travel Fund, Oklahoma State University (2025)**: Awarded travel funding to support conference participation and research dissemination.
- **GPSGA Travel & Research Award, Oklahoma State University (2024 and 2025)**: Awarded for conference and research support. [Link]
- **Honorable Mention, ICT Fest**: Islamic University of Technology (2014)
- **Honorable Mention, NCPC**: Daffodil International University (2014)
- **Champion, ICT Olympiad – CSE Fest**: RUET (2012)

VOLUNTARY SERVICES

- **National High School Programming Contest (NHSPC), Rajshahi**: Volunteer 2016
- **Divisional Mathematical Olympiad, Faridpur**: Math Olympiad Volunteer (MOVer) 2006
- Reviewed research papers for COLING (2024), ICWSM (2026), IJCNN (2024), and PAKDD (2025), NeurIPS (2026), ACM Transactions on Internet Technology (2026): Reviewer

REFERENCES

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